INCH-POUND

MIL-C-50652/9A (AR)
27 September 1993
SUPERSEDING
MIL-C-50652/9 (MU)
15 April 1971

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 10548125

(COUNTER A, RANGE RATE)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. **Scope.** The requirements for procuring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50652.

2. Components and assemblies. The components and assemblies. shall comply with all requirements specified on drawing F10548125, all associated drawings, and with all requirements specified in applicable specifications and standards. The circuit card assembly shall meet the requirements of MIL-C-50652.

3. Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

Beneficial comments (recomendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New dersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end Of this document or by letter.

AMSC N/A FSC 1285

DISTRIBUTION STATEMENT A. Approvedforpublic release; distribution isunlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4. Performance. For the requirements of 4.3 the following conditions shall apply:

a. Logic 0: 0.0 + 0.5 volt direct current (Vdc) level.

b. Logic 1: + 3.5 ± 1.0 Vdc level.

c. Positive pulse: a change in direct current (de) level from 0.0 + 0.5 V to + 3.5 ± 1.0 V and return to 0 V.

d. Negative pulse: A change in dc level from + 3.5 ± 1.0 V to 0.0 + 0.5 V and return to + 3.5 V.

4.1 Input Dower. The input power applied between pins Pl-30, 31 and 32 (+5 V) and pins P1-24, 25 and 26 (+5 V return) shall be 5.0 ± 0.3 Vdc, at 300 milliamperes (mA)8 regulated, with an alternating current (at) ripple less than 50 millivolts (mV) peak-to-peak.

4.2 Input signal. The input signal shall be a positive pulse of 20 ± 0.1 kilohertz (kHz), 3.0 ± 1.0 Vdc amplitude, 0.8 ± 0.1 microseconds (µs) duration and 0.1 µs maximum rise and fall time applied to pin P1-8.

4.3 **OUtpUt** signals.

4.301 **up COU**nt signals. With the input logic levels applied as specified in Table I, Step 1, the output signals shall be square waves with frequencies as specified in Table II. The positive transitions of each flip flop set or reset output shall be coincident with alternate positive transitions of the previous stage.

TABLE I. Input lode levels.

	PI PINS				
STEP	16	53	11	14	51
1	1	0	1	1	1
2	0	1	1	1	1
3	1	0	0	1	1

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE 11. Output Signals.

F-F Set Outputs		F-F Reset Outputs		
P1 Pins	(TP) *	P1 Pins	(TP)*	Frequency
7	(1)	13	(21)	10 kHz
9	(2)	12	(20)	5 kHz
56	(3)	10	(19)	2.5 kHz
37	(4)	21	(18)	1.25 kHz
33	(5)	19	(17)	625 Hz
35	(6)	23	(16)	312.5 Hz
38	(7)	36	(15)	156 Hz
40	(8)	22	(14)	78 Hz
48	(9)	45	(13)	39 Hz

* (TP) = Test Point

4.3.2 Down count signals. With the input logic levels applied as specified in Table I, Step 2, the output signals shall be square waves with frequencies as specified in Table II. The positive transitions of each flip flop set or reset output shall be coincident with alternate negative transitions of the previous stage.

4.3.3 Reset signals. With the input logic levels applied as specified in Table I, Step 3, the output signals shall be logic levels as specified in Table 111.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE XII. Reset output signals.

LOGIC 1		LOGIC 0	
P1 Pins	(TP)	I Pi Pins	(TP)
7	(1)	13	(21)
9	(2)	12	(20)
56	(3)	10	(19)
37	(4)	21	(18)
33	(5)	19	(17)
35	(6)	23	(16)
38 .	(7)	36	(15)
40	(8)	22	(14)
48	(9)	45	(13)

4.3.4 Pulse output signals. With the input logic levels applied sequentially as specified in Table IV, the pulse output signals shall be as specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLB IV. Pulse output signals.

STEP	INPUT PINS 16	53	11	14	51	OUTPUT PIN	OuTPuT (TP)	SIGNAL CHARACTERISTIC 1/
1	0	1	1	1	1	5	(22)	Positive pulse, 100 µs PRF
2	1	1	0	1	1	5	(22)	Logic 0 level
3	0	1	1	1	1	1	(23)	Positive pulse, 200 µs PRF
4	0	1	1	1	1	3	(24)	Positive pulse, 0.4 ms PRF
5	0	1	1	1	1	6	(25)	Positive pulse 0.8 ms PRF
6	0	1	1	1	1	55	(26)	Positive pulse, 1.6 ms PRF
7	0	1	1	1	1	43	(lo)	Negative pulse, 25.6 ms PRF
8	1	0	1	1	1	43	(lo)	Logic 1 level
9	0	1	1	1	1	49	(11)	Positive pulse, 25.6 ms PRF
10	0	1	1	1	1	34	(12)	Logic 1 level
11	1	0	1	1	1	34	(12)	Negative pulse, 25.6 ms PRF
12	1	1	1	0	0	49	(11)	Positive pulse, 50 µs PRF

1/ All pulses shall have a duration of 0.8 ± 0.1 µs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

5.0 Test equipment. The test equipment listed in Table V shall be utilized in the performance of the applicable tests specified in 6.2.

TABLE V, Test equipment list*

ITEM	NOMENCLATURE	DESCRIPTION
1	Power supply	5.0 ± 0.3 Vdc, regulated. Ac ripple: less than 50 mv peak-to-peak Capacity: 300 mA.
2	Pulse generator	20 kHz positive pulse; 3.0 ± 1.0 Vdc amplitude, 0.8 ± 0.1 µs wide. Pulse rise and fall times: less than 0.1 µs.
3	Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less. Sweep period 50 ns per division or less. Input resistance: greater than 8 megohms. Input capacitance: less than 25 pF.
4	Test fixture	See Figure 1. See Table VI for test fixture parts list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

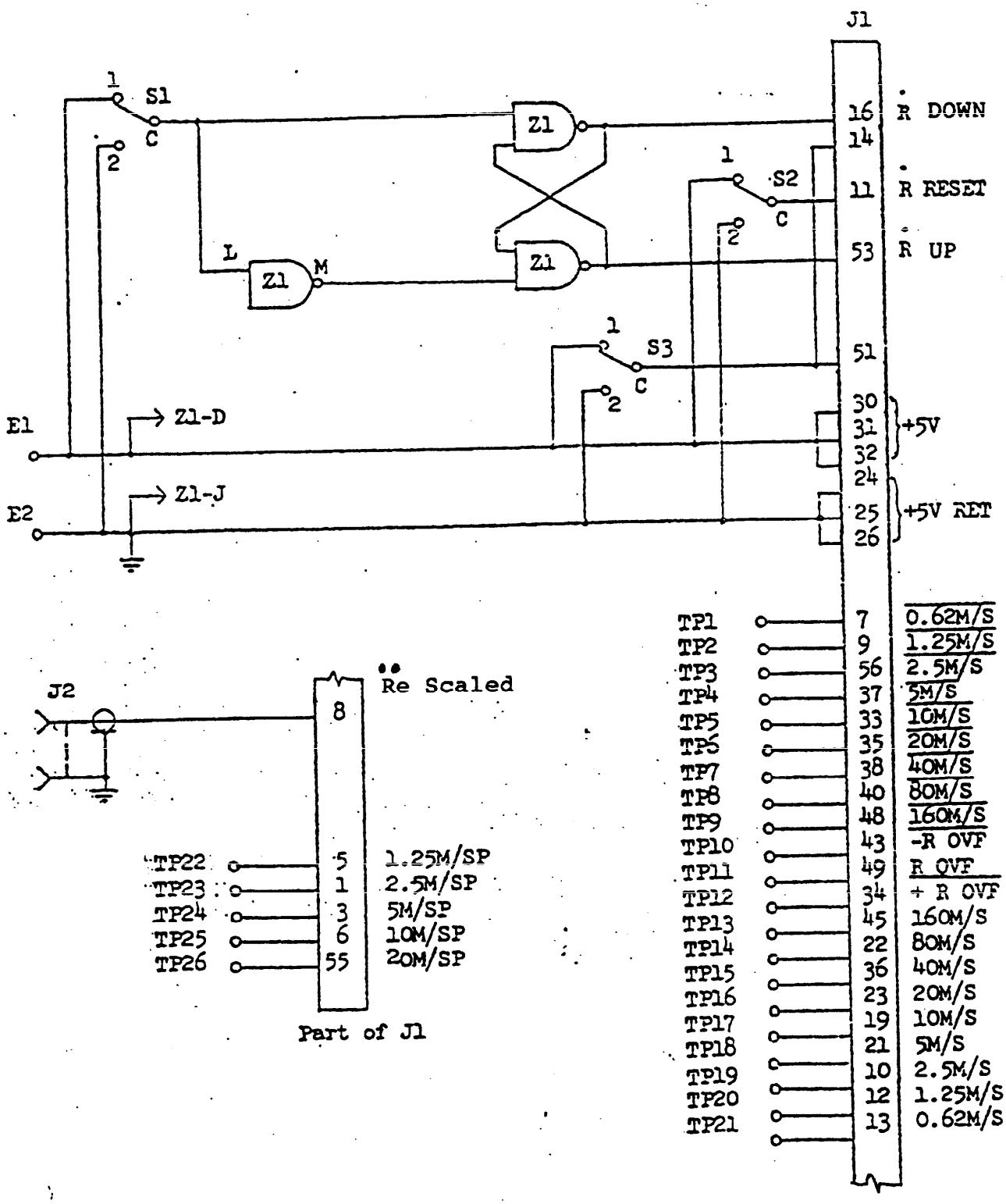


FIGURE 1. TEST FIXTURE SETUP.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE VI. Test fixture parts list.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
El	Binding post, H. H. Smith 257-R or equivalent
E2	Binding post, H. H. Smith 257-B or equivalent
J1	Connector; any of the following or equivalent (1) 5893232-1, Malco Mfg. Co., Inc., Chicago, IL (2) 500946-01, National Connector corp, Minneapolis, MN (3) 07-5422-0198-01, Elco Corp, Willow Grove, PA
J2	Connector, receptacle, UG-25B/U or equivalent
S1	Switch SPDT
S2	Same as S1
S3	Same as S1
TPI thru TP26	Tip jack, H. H. Smith 1501 or equivalent
Z1	Integrated circuit, Quad 2-Input NAND, SN5400 or equivalent

6.0 Functional test.

6.1 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.

6.2 Performance. The selected range registers shall be tested as follows: Connect range register A and test equipment, items 1 through 4, as shown in Figure 2. Pulses and logic levels shall be as specified in 4.0. Use switches S1 through S3 to apply logic levels as follows:

a. S1 - In position 1 applies logic 1 to P1-53 and logic 0 to P1-16, in position 2 applies logic 1 to P1-53 and logic 0 to P1-16.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

b. S2 - In position 1 applies logic 1 to P1-11 and in position 2 applies logic 0 to P1-11.

c. S3 - In position 1 applies logic 1 to P1-14 and 51, in position 2 applies logic 0 to P1-14 and 51.

Use oscilloscope, item 3, to observe input and output signals as specified in 6.2.2 through 6.2.3.4.

6.2.1 Input power. Using power supply, item 1, apply input power as specified in 4.1.

6.2.2 Input signal. Using pulse generator, item 2, set up and apply input signal, as specified in 4.2, to J2 on test fixture, item 4.

6.2.3 Output signals.

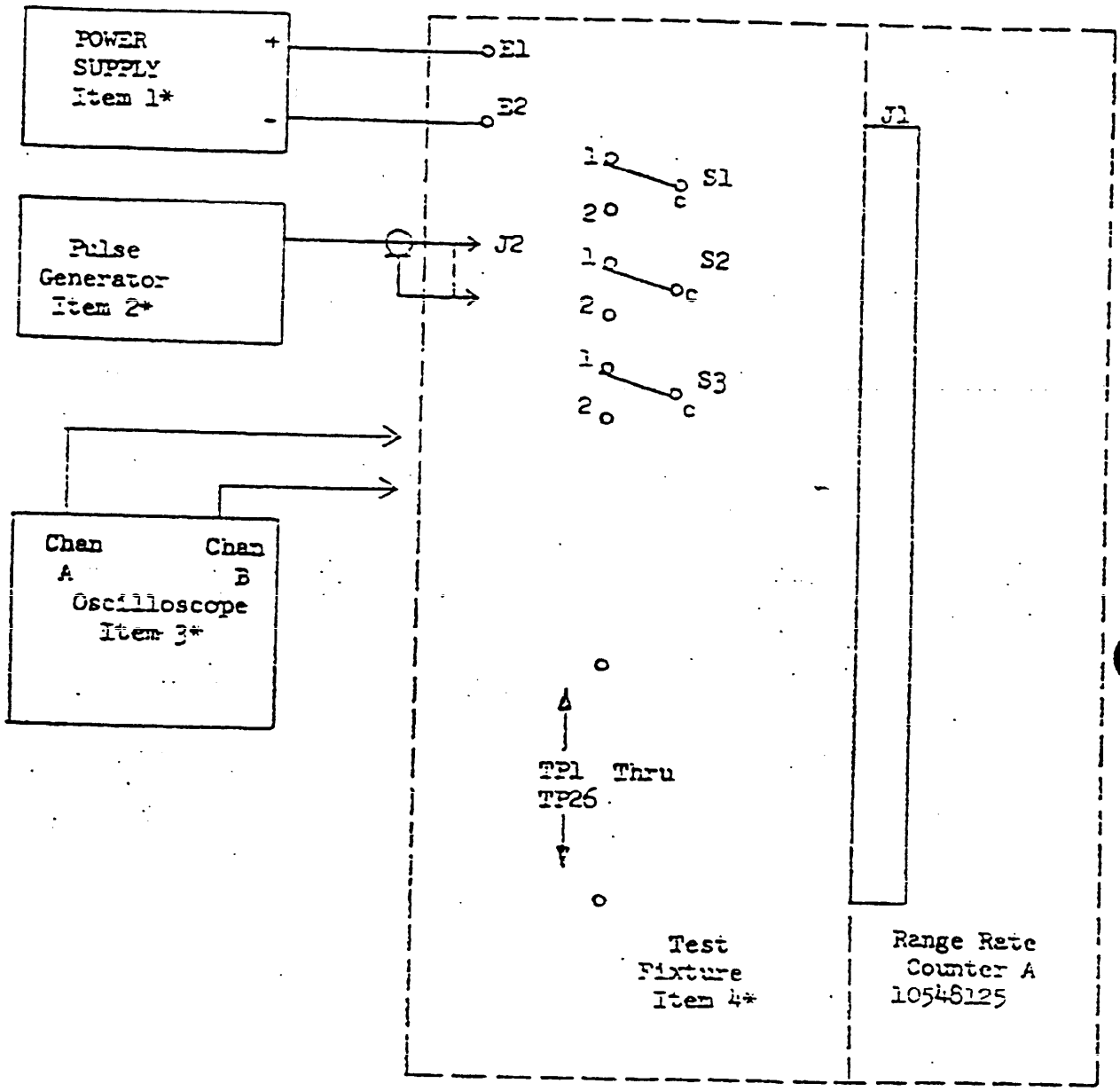
6.2.3.1 Up count signals. Using switches S1 through S3, apply logic levels as specified in Table I, Step 1, and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for phase shift as specified in 4.3.1.

6.2.3.2 Down count signals. Using switches S1 through S3, apply logic levels as specified in Table I, Step 2 and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for being in phase as specified in 4.3.2.

6.2.3.3 Reset signals. Using switches S1 through S3, apply logic levels as specified in Table I, Step 3, and observe the output signals as specified in Table III.

6.2.3.4 Pulse output signals. Using switches S1 through S3, apply logic levels as specified in Table IV and observe the output signals as specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.



*Refer to 5.0

FIGURE 2. Range rate counter A test setup.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

7.0. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A879)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.